**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 266 · Paging, segmentation and virtual memory

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is RAM used for?

Q2. What are remote virtual storage files?

Q3. Why might a process have to wait on memory I/O?

Q4. A 4-row, 3-column zero-indexed array: give its final valid index pair and number of values.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

page — Virtual fixed-size memory block

frame — Physical fixed-size RAM block

page fault — Page absent from RAM

segmentation — Logical variable-length regions

## New learning

Understand — Pages and frames have equal fixed sizes; the page table maps virtual pages to physical frames.

Trace — Divide the virtual address by page size for the page number; use remainder as the offset.

Explain — A page fault requires a non-resident page to be loaded. Variable-sized segmentation has different fragmentation characteristics.

## Worked example

Translate virtual address 520 with page size 128 and page 4 mapped to frame 9.

1. Predict first — 520 div 128=4 remainder 8; frame 9 base=1152, plus 8 gives 1160.

# We Do · Practise with guidance

Virtual address 615 with page size 200, page 3 → frame 7. Calculate physical address.

```
page=3; offset=15; physical=7×200+15=1415
```

Predict first

615 div 200=3 remainder 15. Page 3 maps to frame 7, hence physical address 1415.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Define page, frame, page table and page fault. [4]

### Question 2

Page size 256, VA 770, page 3 mapped to frame 5. Find physical address. [6]

### Question 3

Compare paging and segmentation and explain thrashing. [6]

# Exit Ticket · Answer before checking

Paging changes the offset within the page when it changes frame.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Active program and data.

Q2. Files hosted on network-accessible servers.

Q3. Required page/data is not resident and must be loaded.

Q4. Index (3,2), twelve values.

### You Do pages · Independent questions

Q1. Virtual fixed block [1]; physical fixed block [1]; mapping [1]; access to absent page [1].

Q2. Page=3 [1], offset 2 [2], frame base 1280 [2], final physical address 1282 [1].

Q3. Fixed pages [1] internal fragmentation [1], variable segments [1] external fragmentation [1], frequent page faults [1] slow backing-store transfers [1].

### Exit Ticket · Model answer

False. The offset stays the same.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**